Strategic Endurance under Uncertainty:

A Game-Theoretic Framework for Thinking about the IranU.S. Confrontation

Abstract

This paper offers a conceptual framework for thinking about the Iran–U.S. confrontation as a prolonged strategic competition under uncertainty. It does not claim to provide a political prescription, a predictive model, or a complete explanation of the confrontation. Rather, it proposes a way of organizing the discussion around a central idea: in prolonged confrontations, the outcome is shaped not only by material power, but also by beliefs, perceived endurance, cost management, signaling, domestic stability, and the availability of politically acceptable exit options.

The main argument is that strategic endurance should not be understood simply as the ability to tolerate pressure. Endurance becomes strategically meaningful only when pressure can be transformed into politically manageable costs, when signals of resolve are credible, and when decision-makers preserve enough flexibility to avoid being trapped by their own commitments. In this view, each side observes the other's actions, interprets them as signals, updates its beliefs about the opponent's capacity to continue, and then decides whether continued confrontation remains preferable to compromise or de-escalation.

The purpose of the paper is therefore not to model the Iran–U.S. confrontation in the sense of claiming expert authority over political or security decision-making. Its purpose is more limited: to provide a structured lens through which policymakers, analysts, and informed audiences can think about why costly confrontation may persist, why exit may become politically difficult, and why strategic resilience depends on more than the capacity to suffer.

Keywords: Strategic endurance; uncertainty; Iran–U.S. confrontation; war of attrition; incomplete information; costly signaling; domestic audience costs; face-saving exit; cost-management capacity.

1 Introduction

The Iran–U.S. confrontation remains one of the most persistent strategic rivalries in contemporary international relations. Since 1979, the relationship between the two states has been shaped by sanctions, nuclear diplomacy, regional competition, military coercion, intelligence activity, proxy conflict, and recurring cycles of escalation and restraint. Despite the high costs imposed on both sides, the confrontation has not reached a decisive resolution. Each side continues to absorb costs, signal resolve, manage domestic and international audiences, and assess whether the other side may eventually revise its position.

A conventional reading of this confrontation often begins with the asymmetry of material power. The United States possesses far greater military, economic, and institutional capabilities. Yet material superiority alone has not produced strategic closure. Iran has endured decades of sanctions, diplomatic pressure, regional isolation, and repeated threats of military action while preserving the core elements of its strategic posture. This suggests that the confrontation cannot be understood only through the distribution of material capabilities. It also requires attention to endurance, beliefs, cost management, domestic constraints, and the political feasibility of exit.

This paper draws on the rationalist logic of conflict, according to which costly confrontation may persist when actors possess private information about their capabilities, resolve, or costs, and when they have incentives to misrepresent that information. Fearon's classic account identifies private information and incentives to misrepresent as one of the core rationalist explanations for war and bargaining failure [1].

The paper also draws on the logic of the war of attrition. In a war of attrition, players choose how long to remain in a costly contest, hoping that the opponent will exit first. The central trade-off is between the value of outlasting the opponent and the costs incurred while waiting.

However, the Iran–U.S. confrontation cannot be represented adequately by a simple textbook war of attrition. The confrontation is dynamic, politically complex, and characterized by incomplete information on both sides. Moreover, actions are observed and interpreted as signals. This means that the model should be understood as a multi-stage game with observed actions and incomplete information. In such games, strategies must be sequentially rational in continuation games, and beliefs must be updated by Bayes' rule whenever possible.

This paper does not seek to offer a definitive political or security assessment of the Iran–U.S. confrontation, nor does it claim to substitute for the judgment of policymakers, diplomats, or regional experts. Its aim is more limited and analytical: to provide a structured framework for thinking about why costly confrontations may persist under uncertainty, why de-escalation may become politically difficult, and why strategic resilience depends on more than the capacity to absorb pressure.

The central argument is that prolonged confrontation is not merely a contest over material power; it is also a contest over perceived endurance. Each side observes the other's behavior, interprets signals, updates beliefs, and decides whether continuing the confrontation remains preferable to compromise, restraint, or de-escalation. In this setting, the key question is not simply which side is stronger, but which side can sustain its own form of endurance in a way that appears credible, politically manageable, and strategically purposeful.

2 The Meaning of Endurance for Iran and the United States

In this paper, endurance has different meanings for the two players. The confrontation is not a symmetric contest in which both sides endure the same kind of pain.

For Iran, endurance refers primarily to the capacity to absorb external pressure without allowing that pressure to become internal destabilization. It includes the ability to manage sanctions, preserve domestic order, maintain elite cohesion, protect critical economic functions, stabilize public expectations, and prevent economic pressure from becoming social or political crisis. In this sense, Iranian endurance is closely related to the state's ability to transform external pressure into politically manageable costs.

For the United States, endurance has a different meaning. It does not primarily refer to survival under pressure. Rather, it refers to the ability to sustain pressure on Iran over time without the policy becoming domestically unpopular, militarily overextended, diplomatically isolated, fiscally costly, or strategically uncontrollable. U.S. endurance depends on domestic political support, congressional tolerance, alliance coordination, military readiness, control of escalation risk, clarity of strategic objectives, and the availability of a politically defensible exit strategy.

This distinction is central to the argument. Iran's endurance is mainly about absorbing pressure without internal fracture. U.S. endurance is mainly about sustaining coercive pressure without strategic overextension or domestic political backlash. The two forms of endurance are different, but both are strategically relevant because each side observes the other's behavior and tries to infer whether continued confrontation remains worthwhile.

3 Game Setting: Strategic Endurance as a Multi-Stage War of Attrition under Uncertainty

This section presents the formal setting of the game. The confrontation is modeled as a multi-stage game with observed actions, incomplete information, history-dependent movers, costly signaling, and evolving endurance.

The game is not a simultaneous-move interaction at every stage. At some histories, the United States moves first and Iran observes and responds. At other histories, Iran moves first and the United States responds. In some stages, both players may act simultaneously. Diplomatic messages, ceasefire bargaining, military escalation, restraint, and retaliation can therefore be represented within the same dynamic framework.

The game is denoted by

$$\Gamma = \langle N, \Theta, p, E, T, H, M, A, \Sigma, D, S, g, G, \varphi, B, X, \mu, \tau, u \rangle.$$

The following table summarizes the variables, functions, and primitives of the game.

The table defines the primitives of the game. The structural type

$$\theta_i = (\bar{\rho}_i, \bar{m}_i)$$

is drawn at the beginning of the game and remains fixed. It captures player i's baseline resolve and baseline cost-management capacity. By contrast,

$$e_i^t$$

is the realized endurance state of player i at stage t. Endurance is not fixed. It evolves over time as the confrontation develops. Thus, θ_i is the structural source of endurance, while e_i^t is the current level of endurance at a particular stage.

The public history

$$h^t = (a^0, a^1, \ldots, a^{t-1})$$

records all actions observed before stage t. The active-mover set

$$M(h^t) \subseteq N$$

allows the timing of the game to depend on history. At some histories, only one player moves

Table 1: Components of the Game Γ

Component	**Definition**
N	$\{I, U\}$ — Iran and the United States
Θ	$\theta_i = (\bar{\rho}_i, \bar{m}_i) \in \Theta_i$ — private structural type of player i, where $\bar{\rho}_i$ denotes baseline resolve and $\bar{m}_i$ denotes baseline cost-management capacity
p	$p_i(\theta_i)$ — prior distribution over player i's structural type; the joint prior is $p(\theta) = p_I(\theta_I)p_U(\theta_U)$ under type independence
$\mathcal{E}$	$e_i^t \in \mathcal{E}_i$ — latent endurance state of player i at stage t; initial endurance is determined by the structural type, $e_i^0 = e_i^0(\theta_i)$
T	$t = 0, 1, \ldots, \bar{T}$ — stages of the confrontation
H	$h^t = (a^0, a^1, \ldots, a^{t-1})$ — public history of observed actions before stage t
M	$M(h^t) \subseteq N$ — active mover set after history h^t; the timing of actions may be sequential or simultaneous depending on the history
A	$a_i^t = (\sigma_i^t, d_i^t) \in A_i(h^t) = \Sigma_i(h^t) \times \{C, E\}$ — complete action of player i after history h^t, if $i \in M(h^t)$
Σ	$\sigma_i^t \in \Sigma_i(h^t)$ — observable signal component of a_i^t: military posture, diplomatic message, restraint, escalation, public commitment, or performance under pressure
D	$d_i^t \in \{C, E\}$ — decision component of a_i^t: continue (C) or exit/de-escalate/compromise/change strategy (E)
S	$s_i = \{s_i^t\}_{t=0}^{\bar{T}}$, where $s_i^t(\cdot \mid h^t, \theta_i, e_i^t) \in \Delta(A_i(h^t))$ — behavior strategy assigning a probability distribution over feasible complete actions
G	$G_i(t; \theta_i, e_i^t, h^t)$ — accumulated effective cost up to stage t; the flow cost is $g_i(t, h^t; \theta_i, e_i^t) = \dfrac{c_i(t, h^t) + a_i^{\text{aud}}(t, h^t) + r_i(t, h^t)}{m_i^t}$
$c_i(\cdot)$	$c_i(t, h^t)$ — material and economic cost of continuing the confrontation after history h^t
$a_i^{\text{aud}}(\cdot)$	$a_i^{\text{aud}}(t, h^t)$ — domestic audience cost: public dissatisfaction, elite pressure, protest risk, electoral pressure, legitimacy cost, or reputational cost
$r_i(\cdot)$	$r_i(t, h^t)$ — escalation-risk cost: expected cost of military escalation, miscalculation, retaliation, regional spillover, or loss of crisis control
m_i^t	$m_i^t = m_i(\theta_i, e_i^t, h^t) > 0$ — current cost-management capacity; the ability to absorb, distribute, contain, or politically manage raw costs
B	$B_i(e_i^t, \theta_i)$ — value of outlasting the opponent; increasing in current endurance and baseline resolve
X	$X_i(t, h^t)$ — dynamic exit value; value of compromise, de-escalation, ceasefire, or strategic adjustment after history h^t
u	u_i — payoff: value of outlasting minus effective and signaling costs, or exit value minus effective and signaling costs
μ	$\mu_j^t(\theta_i, e_i^t \mid h^t)$ — posterior belief of player j about player i's structural type and current endurance after history h^t

and the other observes and responds later. At other histories, both players choose actions simultaneously. This formulation allows military attacks and responses, diplomatic messages, ceasefire bargaining, restraint, escalation, and simultaneous strategic choices to be represented within one framework.

4 Actions, Signals, and Strategies

The complete action of player i at stage t is

$$a_i^t = (\sigma_i^t, d_i^t).$$

The component σ_i^t is the observable signal component of the action. It may include military posture, diplomatic messaging, public commitment, restraint, escalation, or observable performance under pressure.

The component d_i^t is the continuation decision. The player either continues the confrontation, denoted by C, or exits, de-escalates, compromises, accepts a ceasefire, or changes strategy, denoted by E.

The signal σ_i^t should not be understood as a separate action from a_i^t. Rather, it is the informational component of the complete action. The same action can affect the material state of the confrontation and also convey information about the player's resolve, endurance, and willingness to bear costs.

A behavior strategy for player i is

$$s_i = \{s_i^t\}_{t=0}^{\bar{T}},$$

where

$$s_i^t(\cdot \mid h^t, \theta_i, e_i^t) \in \Delta(A_i(h^t)).$$

Thus, after every public history, structural type, and current endurance state, the strategy assigns a probability distribution over feasible complete actions. For a particular action a_i^t, the probability of choosing it is written as

$$s_i^t(a_i^t \mid h^t, \theta_i, e_i^t).$$

In a pure-strategy version of the model, this distribution assigns probability one to a single complete action.

5 Effective Cost of Continuing

The effective flow cost of continuing the confrontation is

$$g_i(t, h^t; \theta_i, e_i^t) = \frac{c_i(t, h^t) + a_i^{aud}(t, h^t) + r_i(t, h^t)}{m_i^t}.$$

Here, $c_i(t, h^t)$ denotes the material and economic cost of continuing the confrontation. For Iran, this may include sanctions pressure, disruption of oil revenues, exchange-rate pressure, fiscal stress, or infrastructure vulnerability. For the United States, it may include military deployment costs, alliance-management costs, fiscal burdens, or economic consequences of regional instability.

The term $a_i^{aud}(t, h^t)$ denotes domestic audience cost. It captures the internal political and social cost of continuing, escalating, or retreating. For Iran, this may include public dissatisfaction, protest risk, elite fragmentation, legitimacy costs, or social pressure caused by economic instability. For the United States, it may include public opposition, congressional pressure, electoral costs, media criticism, or dissatisfaction among allies and domestic constituencies.

The term $r_i(t, h^t)$ denotes escalation-risk cost. It captures the expected cost associated with the possibility that the confrontation becomes more dangerous. This includes the risk of military escalation, retaliation, miscalculation, regional spillover, attacks on forces or infrastructure, or loss of control over the crisis.

The denominator m_i^t denotes current cost-management capacity:

$$m_i^t = m_i(\theta_i, e_i^t, h^t) > 0.$$

It measures player i's ability at stage t to absorb, distribute, contain, or politically manage raw costs. A higher m_i^t reduces the effective burden of material, domestic, and escalation costs. Thus, the same level of raw pressure can generate different effective costs depending on the player's ability to manage that pressure.

The accumulated effective cost up to stage t is

$$G_i(t; \theta_i, e_i^t, h^t) = \sum_{s=0}^{t} g_i(s, h^s; \theta_i, e_i^s).$$

This term measures the total effective burden borne by player i up to stage t.

6 Costly Signaling

Signals are costly. The signaling cost is denoted by

$$\varphi_i(\sigma_i^s; \theta_i, e_i^s, h^s).$$

This is the cost borne by player i to produce, maintain, and make credible the observable signal σ_i^s at stage s. It is not merely the cost of revealing information. Rather, it includes the real resource, political, opportunity, escalation, and commitment costs associated with the action that produces the signal.

For example, if σ_U^s is a high military posture by the United States, then $\varphi_U(\sigma_U^s; \theta_U, e_U^s, h^s)$ may include deployment costs, logistical costs, readiness costs, alliance-coordination costs, domestic political costs, escalation exposure, and the loss of diplomatic flexibility. Similarly, if σ_I^s is a signal of resilience by Iran, the cost may include maintaining domestic stability, preserving economic continuity, sustaining military readiness, and absorbing political pressure.

In foreign-policy signaling theory, costly signals are often discussed through the distinction between sunk-cost signals and tying-hands signals. Fearon distinguishes between signals that impose costs up front and signals that create future political or reputational costs if the leader fails to follow through.

The signaling cost can be decomposed as

$$\varphi_i(\sigma_i^s;\theta_i,e_i^s,h^s) = \varphi_i^{prod}(\sigma_i^s;\theta_i,e_i^s,h^s) + \varphi_i^{commit}(\sigma_i^s;\theta_i,e_i^s,h^s),$$

where φ_i^{prod} is the production or maintenance cost of the signal, and φ_i^{commit} is the commitment or audience cost created by making the signal publicly observable.

We assume that

$$\varphi_i(\sigma_i^s;\theta_i,e_i^s,h^s) \geq 0, \qquad \frac{\partial \varphi_i}{\partial \sigma_i^s} > 0.$$

Thus, signaling is costly, and stronger signals are more costly.

We also assume that

$$\frac{\partial \varphi_i}{\partial e_i^s} < 0, \qquad \frac{\partial \varphi_i}{\partial \bar{\rho}_i} < 0, \qquad \frac{\partial \varphi_i}{\partial \bar{m}_i} < 0.$$

This means that the same signal is less costly for players with higher current endurance, stronger baseline resolve, or greater baseline cost-management capacity.

Finally, to capture credibility, we impose a single-crossing-type condition:

$$\frac{\partial^2 \varphi_i}{\partial \sigma_i^s \, \partial e_i^s} < 0.$$

Equivalently, with respect to baseline resolve,

$$\frac{\partial^2 \varphi_i}{\partial \sigma_i^s \, \partial \bar{\rho}_i} < 0.$$

This means that the marginal cost of increasing signal strength is lower for high-endurance or high-resolve players than for low-endurance or low-resolve players. As a result, strong signals can become informative because weak types can imitate them only at a higher cost.

7 Value of Outlasting and Value of Exit

The value of outlasting the opponent is denoted by

$$B_i(e_i^t, \theta_i).$$

This is the strategic benefit obtained if player i remains in the confrontation until the opponent exits, de-escalates, or compromises. The value of outlasting is assumed to increase with current endurance and baseline resolve:

$$\frac{\partial B_i}{\partial e_i^t} > 0, \qquad \frac{\partial B_i}{\partial \bar{\rho}_i} > 0.$$

Thus, a player with greater endurance or stronger baseline resolve attaches greater value to remaining in the confrontation until the opponent changes course.

The exit value is denoted by

$$X_i(t, h^t).$$

Exit does not necessarily mean defeat. It may mean compromise, de-escalation, ceasefire, diplomatic adjustment, or strategic change of course. The value of exit is dynamic because the political and strategic value of leaving depends on the public history. It may increase when a face-saving path becomes available, when third-party mediation is credible, or when the opponent's commitments are reliable. It may decrease when domestic audience costs make compromise politically difficult or humiliating.

This is one of the key insights of the framework: a player may continue not because victory is likely, but because exit is politically unavailable at that moment.

8 Beliefs and Updating

Player j's posterior belief about player i's structural type and current endurance is

$$\mu_j^t(\theta_i, e_i^t \mid h^t).$$

Since endurance is not directly observed, each player infers the opponent's endurance from observed actions, signals, and performance under pressure. Beliefs are updated by Bayes' rule whenever the observed history has positive probability.

This is why the confrontation is not only a contest of material pressure but also a contest of interpretation. Each action changes not only the state of the confrontation but also the opponent's belief about how long one can continue.

9 Utility Function

For each player $i \in \{I, U\}$, let $j \neq i$. The stopping time of player i is defined as

$$\tau_i = \inf\{t \in \{0, 1, \ldots, \bar{T}\} : d_i^t = E\}.$$

If player i never exits before the terminal stage, then

$$\tau_i = \bar{T}.$$

The realized utility of player i is

$$u_i = \begin{cases} B_i(e_i^{\tau_j}, \theta_i) - G_i(\tau_j; \theta_i, e_i^{\tau_j}, h^{\tau_j}) - \displaystyle\sum_{s=0}^{\tau_j} \varphi_i(\sigma_i^s; \theta_i, e_i^s, h^s), & \text{if } \tau_i > \tau_j, \\ X_i(\tau_i, h^{\tau_i}) - G_i(\tau_i; \theta_i, e_i^{\tau_i}, h^{\tau_i}) - \displaystyle\sum_{s=0}^{\tau_i} \varphi_i(\sigma_i^s; \theta_i, e_i^s, h^s), & \text{if } \tau_i \le \tau_j. \end{cases}$$

The first case corresponds to outlasting the opponent. Player i receives the strategic value B_i, but pays accumulated effective costs and signaling costs until the opponent exits.

The second case corresponds to exiting first or exiting simultaneously with the opponent. Player i receives the dynamic exit value X_i, but still bears the effective and signaling costs accumulated up to its own exit.

Therefore, the central decision at each stage is whether the expected value of continuing exceeds the value of exit. The confrontation persists as long as both players believe that continuing remains preferable to exiting, de-escalating, compromising, or changing strategy.

10 Credibility–Flexibility Trade-Off

A central implication of the model is the credibility–flexibility trade-off.

A stronger signal may increase credibility. If a player publicly commits to a position, raises military posture, absorbs costs without visible instability, or refuses immediate compromise, the opponent may infer that the player has high endurance. This can strengthen deterrence or bargaining leverage.

However, stronger signals may also reduce flexibility. A public commitment may increase domestic audience costs if the player later compromises. A military posture may raise escalation risk. A maximalist diplomatic position may make face-saving exit more difficult.

Thus, the same signal can have two opposing effects:

$$\text{Stronger signal} \Rightarrow \text{higher credibility}$$

but also

$$\text{Stronger signal} \Rightarrow \text{lower flexibility.}$$

The optimal strategy is not necessarily to send the strongest possible signal. A player must choose a signal strong enough to be credible but not so strong that it traps the player in an inflexible position.

This is especially important in prolonged confrontations. If a player sacrifices too much flexibility in order to appear resolute, it may later continue the confrontation not because continuation is strategically optimal, but because exit has become politically too costly.

11 Strategic Interpretation

The model implies that strategic endurance is not equivalent to raw pain tolerance. A player may be willing to suffer, but if suffering produces domestic instability, elite fragmentation, economic crisis, or political backlash, its effective endurance declines.

Strategic endurance has three elements: the capacity to absorb pressure, the capacity to manage pressure politically, and the capacity to signal endurance credibly without losing exit flexibility.

For Iran, this means that endurance depends not only on willingness to resist pressure, but also on the ability to prevent pressure from becoming internal destabilization. For the United States, endurance depends not only on the ability to apply pressure, but also on the ability to sustain that pressure without domestic fatigue, alliance erosion, military overextension, or uncontrolled escalation.

The confrontation is therefore a contest over beliefs. Each player asks: can the opponent continue? Is the opponent's signal credible? Are the opponent's costs becoming politically unmanageable? Is exit becoming more attractive or less available? These beliefs determine whether each player sees continued confrontation as worthwhile.

12 Policy Implications

The framework has several policy implications.

First, cost management is strategically central. Pressure does not automatically produce strategic advantage. Its effect depends on how it is absorbed and managed. If a state transforms external pressure into manageable political cost, its endurance remains high. If it fails to do so, pressure can become domestic vulnerability.

Second, signaling must be credible but not self-trapping. Signals that are too weak may fail to convince the opponent. Signals that are too strong may increase domestic audience costs and reduce the political feasibility of exit. Effective strategy requires balancing credibility and flexibility.

Third, exit must be constructed. In prolonged confrontations, exit is not simply an available outside option. It must often be made politically acceptable through sequencing, mediation, reciprocal steps, face-saving language, or credible commitments. A player may continue not because it expects victory, but because no acceptable exit path exists.

Fourth, domestic stability is part of strategic endurance. In this framework, internal political and economic management is not separate from external confrontation. It directly affects the opponent's beliefs about endurance.

Fifth, third parties can matter because they may change the value of exit. Mediators, allies, regional actors, and international institutions can provide mechanisms that make de-escalation less humiliating or more credible.

13 Conclusion

This paper has proposed a conceptual framework for thinking about the Iran–U.S. confrontation as a prolonged strategic competition under uncertainty. The model builds on the logic of the war of attrition, but modifies it to include incomplete information, observed actions, costly signaling, evolving endurance, dynamic exit values, and history-dependent move structures.

The central claim is that strategic endurance is not simply the capacity to tolerate pressure. Endurance becomes strategically meaningful only when pressure can be transformed into politically manageable cost, when signals of resolve are credible, and when the player preserves a feasible path to exit.

In the Iran–U.S. confrontation, Iran's endurance is mainly about absorbing external pressure without internal destabilization. U.S. endurance is mainly about sustaining coercive pressure without domestic fatigue, diplomatic isolation, military overextension, or uncontrolled escalation. These two forms of endurance are different, but both are strategically relevant because each side continuously observes, interprets, and updates beliefs about the other.

The model therefore shifts attention from material power alone to the interaction among costs, beliefs, signals, and exit options. A player does not win simply by suffering longer. Nor does a player win simply by imposing costs. The strategic advantage belongs to the side that can manage costs, sustain credible signals, preserve domestic stability, and maintain enough flexibility to avoid being trapped by its own commitments.

In this sense, strategic resilience is not merely the ability to endure pain. It is the ability to endure pressure in a politically sustainable, strategically credible, and diplomatically usable way.

References

[1] Fearon, James D. 1995. "Rationalist Explanations for War." *International Organization* 49(3): 379–414.